Forecasting and Macroeconomic Analytics

ECON 522

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ARMA models

Univariate Time Series Modeling

- Purpose: to provide a simple description of the basic features/characteristics of a single time series in terms of its own past and errors/innovations to the process in the past.
- Only an approximation to the true process (actual process is simple governed by its interactions with many other processes)
- Although an approximation we still need knowledge about
  - It is stationary or non-stationary?
  - What is the autocorrelation pattern of the series?
- Also useful as a forecasting device given no theory model is available.
- No much knowledge is gained with respect to economic theory
- Important in the way multivariate models are formulated

Moving Average Model (MA)

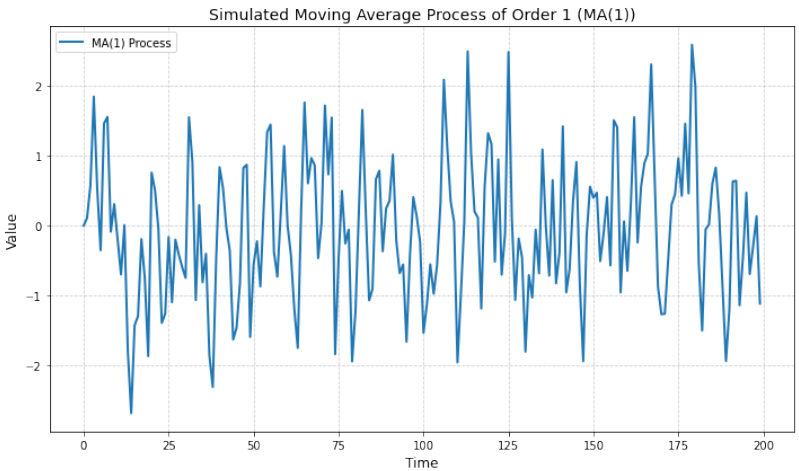
Let $\{\varepsilon_t\}$ be a $WN\left(0, \sigma^2\right)$

- MA(1) $Y_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}$
- MA(2) $Y_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2}$
- MA(q) $Y_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \ldots + \theta_q\varepsilon_{t-q}$

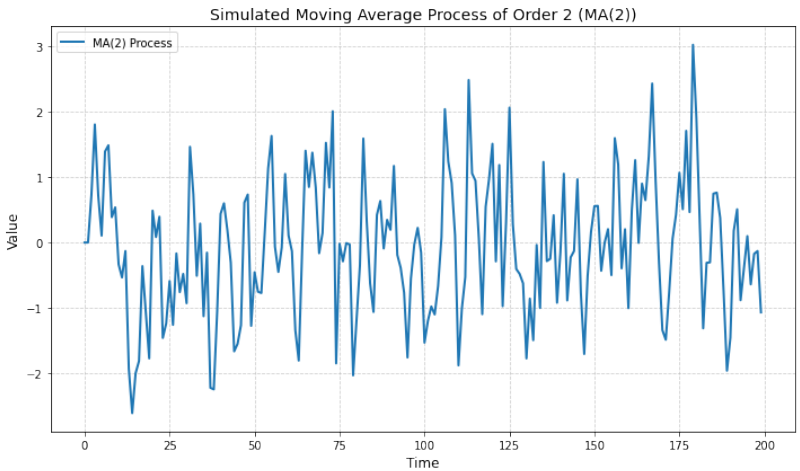
Recall the Wold theorem: any stationary Time Series can be represented by a MA of infinite order.

If Y is a MA is it stationary? For $MA(\infty)$ we need $\sum_{j=0}^{\infty} |\theta_j| < \infty$ What if q is finite?

Simulated MA



(a) MA(1)

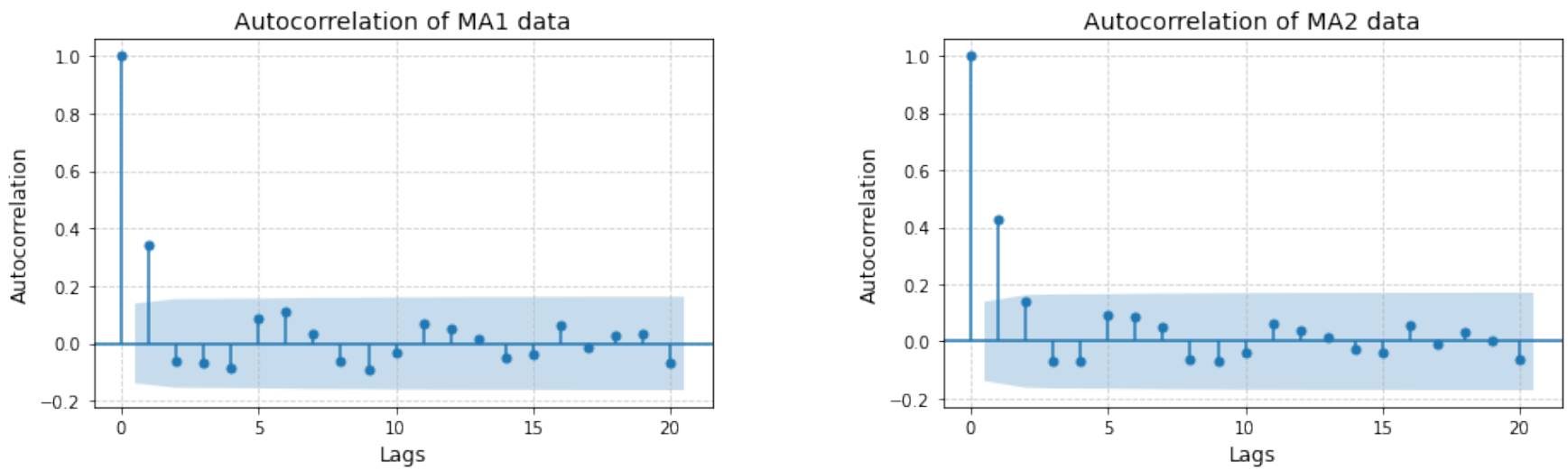


(b) MA(2)

Figure 1: Simulated MA data

ACF of Simulated MA

☆ listen recording



(a) MA(1) (b) MA(2)

Figure 2: ACF of Simulated MA data

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$

$$VAR(y_t) = 1 + \theta_1^2 + \theta_2^2$$

$$\gamma(1) = E[(\varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2})(\varepsilon_{t-1} + \theta_1 \varepsilon_{t-2} + \theta_2 \varepsilon_{t-3})]$$

$$\gamma(2) = E[(\varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2})(\varepsilon_{t-2} + \theta_1 \varepsilon_{t-3} + \theta_2 \varepsilon_{t-4})]$$

$$\gamma(3) = E[(\varepsilon_t \ \varepsilon_{t-1} \ \varepsilon_{t-2})(\varepsilon_{t-3} \ \varepsilon_{t-4} \ \varepsilon_{t-5})]$$

MA

★ listen here

ACF

drops to zero after "q" lags

It goes down slowly for moving avg.

Solve for MA(1)

Let's see why the ACF and PACF look that way. Consider the MA(1) where ε_t is WN with mean zero and usual variance

$$Y_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}$$

Compute the following:

1. $E\left(Y_t\right) =$
2. $VAR(Y_t) = E\left(Y_t - \mu\right)^2 = E\left(\varepsilon_t + \theta\varepsilon_{t-1}\right)^2 =$
3. $\gamma(1) = E\left(Y_t - \mu\right)\left(Y_{t-1} - \mu\right) =$
4. $\gamma(2) =$
5. Is it stationary? Ergodic?
6. Suppose $\theta = 0.5$, graph the ACF

You can do the same for any order MA(q). It will always be stationary as long as q is finite. If of infinite order you need the summabilty condition $\sum_{j=0}^{\infty} \psi_j^2 < \infty$.

$\varepsilon_t \sim WN(0, \sigma^2)$

$VAR(\varepsilon_t) = \sigma^2 = E[\varepsilon_t^2]$

$y_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}$

$$E[y_t] = \underset{\mu}{\underset{\downarrow}{\mu}} + \underset{0}{\underset{\downarrow}{E[\varepsilon_t}} + \underset{0}{\underset{\downarrow}{\theta\varepsilon_{t-1}}}]$$

* epsilon$_t$ is stationary so variance stays the same no matter what t is

$E[y_t] = \mu$

$$VAR(y_t) = E(y_t - \mu)^2 =$$

$$E[\varepsilon_t + \theta\varepsilon_{t-1})^2 =$$

$$E(\varepsilon_t^2 + \theta^2\varepsilon_{t-1}^2 + 2\theta\varepsilon_t\varepsilon_{t-1}) =$$

$$E(\varepsilon_t^2) + \theta^2 E[\varepsilon_{t-1}^2) + 2\theta \underset{=0}{E(\varepsilon_t \cdot \varepsilon_{t-1})}$$

$$= \sigma^2 + \theta^2\sigma^2 +$$

not a function of time
a function of order

$$y_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}$$

$$E[y_t] = \mu$$

$$\gamma(1) = E[(y_t - \mu)(y_{t-1} - \mu)]$$

$$= E[(\varepsilon_t + \theta\varepsilon_{t-1})(\varepsilon_{t-1} + \theta\varepsilon_{t-2})] =$$

$$= E(\underbrace{\cancel{\varepsilon_t\varepsilon_{t-1}}}_{0} + \theta\varepsilon_{t-1}^2 + \underbrace{\cancel{\theta\varepsilon_t\varepsilon_{t-2}}}_{0} + \underbrace{\cancel{\theta^2\varepsilon_{t-1}\varepsilon_{t-2}}}_{0})$$

carry expectation through

$$\underbrace{\theta E[\varepsilon_{t-1}^2]}_{\sigma^2} + 0 + 0$$

$$\gamma(1) = \theta \cdot \sigma^2$$

$$\rho(1) = ACF(1) = \frac{\gamma(1)}{\gamma(0)} = \frac{\theta \cdot \cancel{\sigma^2}}{(1+\theta^2)\cancel{\sigma^2}} = \frac{\theta}{1+\theta^2}$$

if $Y_t = 3 + \varepsilon_t + 0.4\varepsilon_{t-1}$,

then $\rho(1) = ACF(1) = \frac{0.4}{1+0.4^2}$

$$\gamma(2) = E(\underline{\varepsilon_t} + \theta\underline{\varepsilon_{t-1}})(\underline{\varepsilon_{t-2}} + \theta\underline{\varepsilon_{t-3}}) = 0$$

$\gamma(3) = 0$ . . . these are all products of things that are of different timing

Moving average is always stationary

ACF and PACF for MA

Autocorrelation and partial autocorrelation for MA processes have a very specific shape:

- ACF $\rightarrow$ it is different than zero for $j \leq q$ and zero otherwise
- PACF $\rightarrow$ decays or oscillates slowly

We will see that AR models have a different pattern. This prompted Box and Jenkins to suggest to look at ACF and PACF to decide if we should use a MA or AR model for the data and how many lags to include. (outdated method but it may be still useful to look sometime to have an idea)

AR Models

Autoregressive models are the most used models.

- AR(1)

$$AR(1) : y_t = \rho y_{t-1} + \varepsilon_t, |\rho| < 1$$
$$(1 - \rho L) y_t = \varepsilon_t$$

- AR(2)

$$AR(1) : y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \varepsilon_t$$
$$(1 - \rho_1 L - \rho_2 L^2) y_t = \varepsilon_t$$

- AR(p)

$$AR(1) : y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t$$
$$(1 - \rho_1 L - \rho_2 L^2 - \ldots - \rho_p L^p) y_t = \varepsilon_t$$

$$\boxed{y_t = \rho y_{t-1} + \varepsilon_t}$$

$$y_t - \rho y_{t-1} = \varepsilon_t$$

$$(1 - \rho L) y_t = \varepsilon_t$$

$$y_t = \rho y_{t-1} + \varepsilon_t$$

$$y_t = \rho(\rho y_{t-2} + \varepsilon_{t-1}) + \varepsilon_t$$

$$= \varepsilon_t + \rho \varepsilon_{t-1} + \rho^2 y_{t-2}$$

$$= \varepsilon_t + \rho\varepsilon_{t-1} + \rho^2\varepsilon_{t-2} + \dots$$

$$\boxed{y_t = \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-1}}$$

MA(∞) MA(1)

$\updownarrow$ $\updownarrow$

AR(1) AR(2)

If you have finite # of observations then you want to estimate AR(1)

* Two alternative specifications of ways to deal wit y_t

Example:

$$y_t = \mu + p\, y_{t-1} + \varepsilon_t$$

can be written as:

$$y_t = \sum_{j=0}^{\infty} p^j \mu + \sum_{j=0}^{\infty} p^j \varepsilon_{t-j}$$

$\hookrightarrow E[\ ] \Rightarrow 0$

$$E[y_t] = \sum_{j=0}^{\infty} p^j \mu = (1 + p + p^2 + \dots)\, \mu$$

$\hookrightarrow$ if $|p| < 1 \rightarrow \boxed{\frac{1}{1-p} \cdot \mu}$

converges to

mean of y is finite only if p is less than 1, if p is more, then it explodes.

So this is a condition that allows mean to be finite

$$VAR(y_t) = \mathcal{E}(\varepsilon_t + \rho\varepsilon_{t-1} + \rho^2\varepsilon_{t-1} + \dots)^2$$

$$= \sigma^2 + \rho^2\sigma^2 + \rho^4\sigma^2 + \rho^6\sigma^2 + \dots)$$

$$(1 + \rho^2 + \rho^4 + \rho^6 \dots)\sigma^2$$

this is also an infinte sume ... $= \boxed{\frac{1}{1-\rho^2} \cdot \sigma^2}$

$$\gamma(1) = E(\varepsilon_t + \rho\varepsilon_{t-1} + \rho^2\varepsilon_{t-2} + \dots)(\varepsilon_{t-1} + \rho\varepsilon_{t-2} + \rho^2\varepsilon_{t-3} + \dots)$$

all zero except

all ones that have same timing are not zero

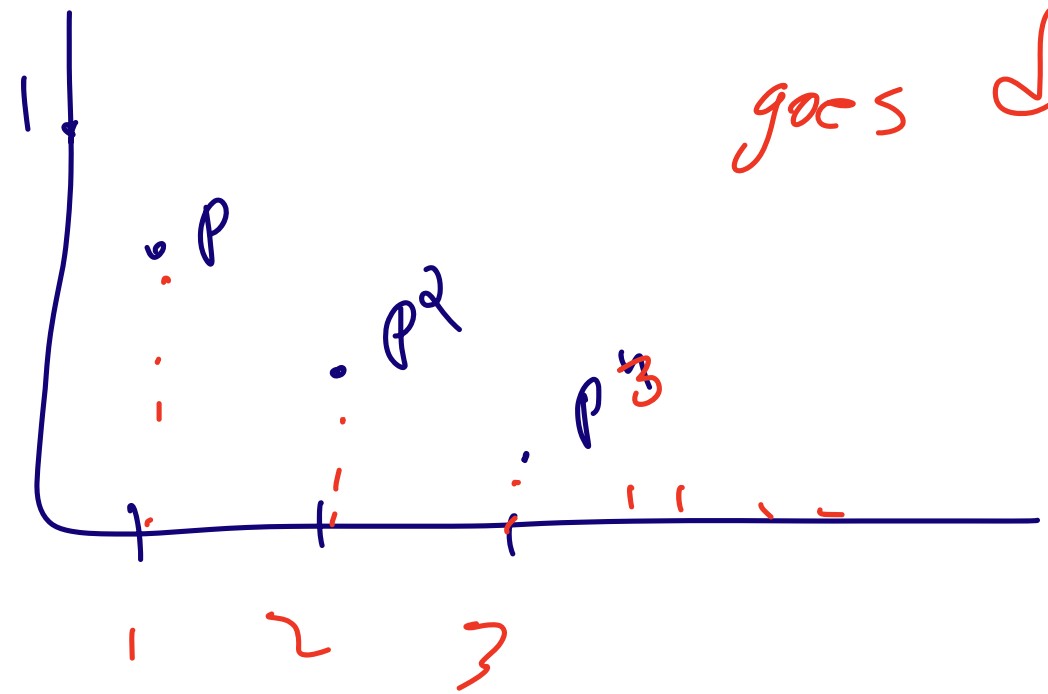
$$= \frac{\rho}{1-\rho^2} + \sigma^2$$

$$\rho(1) = \frac{\cancel{\frac{\rho}{1-\rho^2}} \cdot \cancel{\sigma^2}}{\cancel{\frac{1}{1-\rho^2}} \cdot \cancel{\sigma^2}} = \rho$$

$$\rho(2) = \rho^2$$

$$\rho(3) = \rho^3$$

ACF



goes down slowly

in moving average, ACF goes down slowly

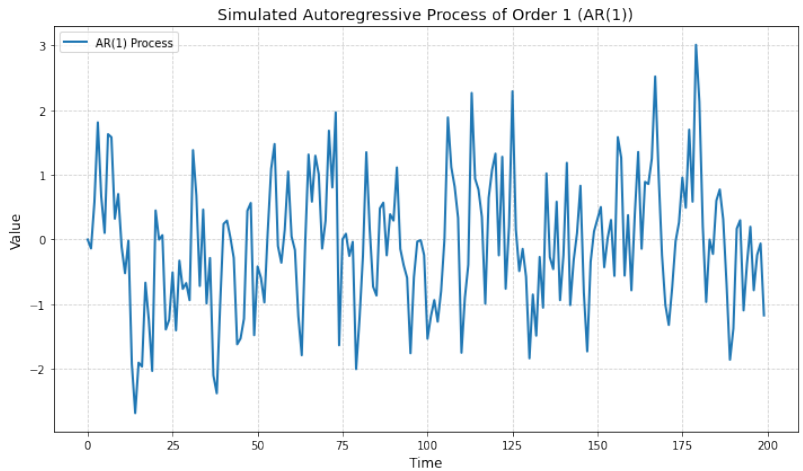
in AR, it goes down to zero really fast

$$y_t = \mu + \rho y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$$

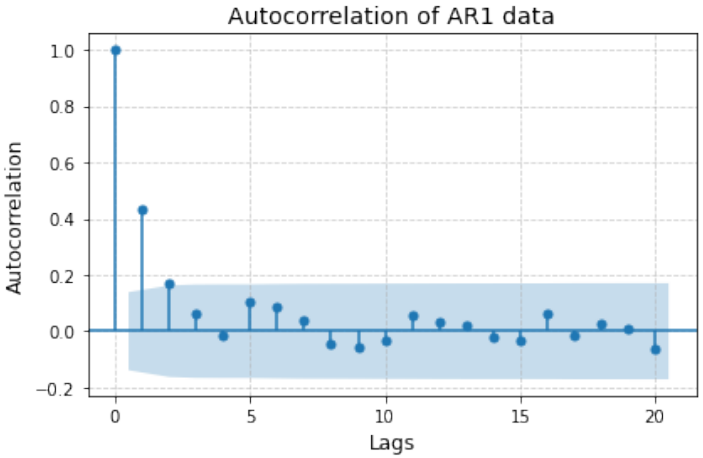
ARMA(1, 1)

Being

Simulated AR(1),$\rho = 0.5$



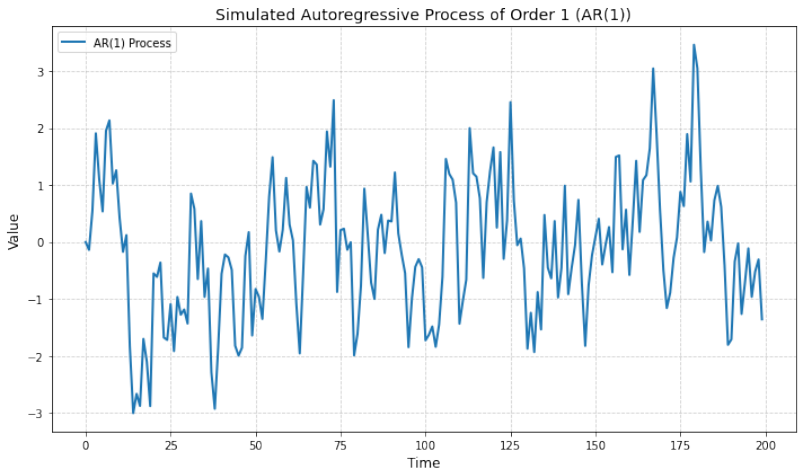
(a) AR(1), $\rho = 0.5$



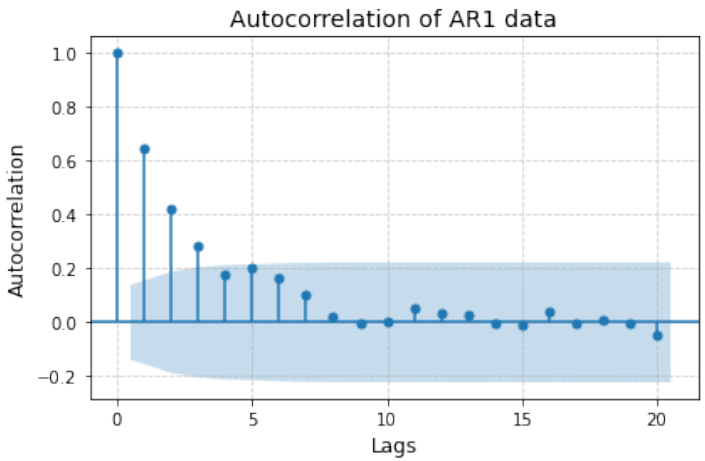
(b) ACF , $\rho = 0.5$

Figure 3: Simulated AR data

Simulated AR(1), $\rho = 0.7$



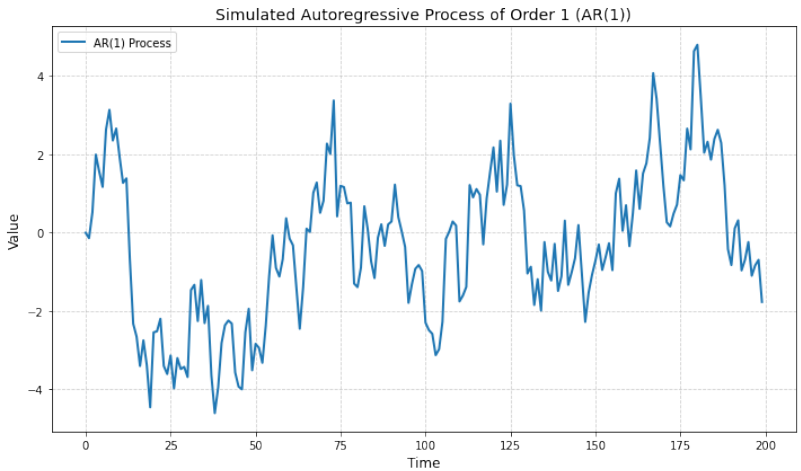
(a) AR(1), $\rho = 0.7$



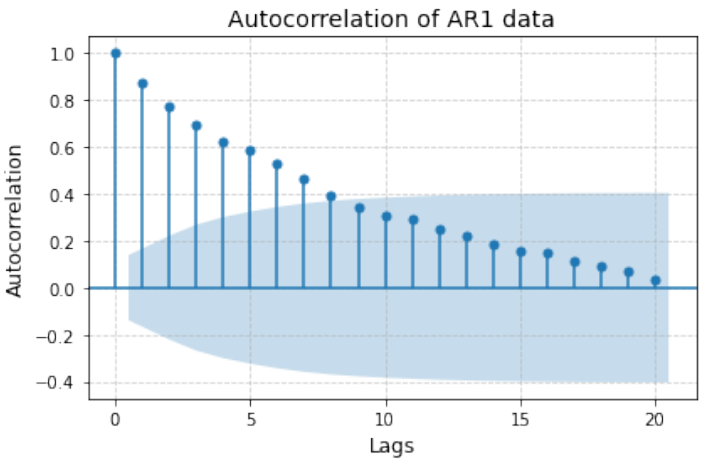
(b) ACF , $\rho = 0.7$

Figure 4: Simulated AR data

Simulated AR(1), $\rho = 0.9$



(a) AR(1), $\rho = 0.9$



(b) ACF , $\rho = 0.9$

Figure 5: Simulated AR data

AR to MA

Notice that we can always write $AR(1)$ (with $|\rho| < 1$) as a $MA(\infty)$:

$$y_t = \sum_{\mathrm{j}=0}^{\infty} \rho^j e_{t-j}$$

You can this formula by repeatedly using the definition of AR(1) or by remembering how to invert a polynomial in the lag operator :

$$\begin{aligned} y_t = \rho y_{t-1} + e_t = \rho \left(\rho y_{t-2} + e_{t-1}\right) + e_t = \\ \ldots = \textstyle\sum_{\mathrm{J}=0}^{k-1} \rho^j e_{t-j} + \rho^k y_{t-k} \end{aligned}$$

and noticing that $\rho^k y_{t-k} \to 0$ as $k \to \infty$.

This is going to be true for all ARMA models. You can go from one to the other.

Solve for AR(1)

It is easier to computer mean and variance from the MA representation: $y_t = \sum_{i=0}^{\infty} \rho^i e_i$

$$E\left(Y_t\right) = \frac{c}{1-\rho} = \mu$$

$$VAR(Y_t) = \gamma(0) = \left(1 + \rho^2 + \rho^4 + \ldots\right)\sigma^2 = \frac{1}{1-\rho^2}\sigma^2$$

$$\gamma(s) = \text{cov}\left(\sum_{i=0}^{\infty} \rho^i e_{t-i}, \sum_{j=0}^{\infty} \rho^j e_{t+k-j}\right) = \sum_{i,j:i=j-k}^{\infty} \rho^{i+j}\sigma^2 = \frac{\rho^k \sigma^2}{1-\rho^2}$$

$$\rho(s) = \frac{\gamma(s)}{\gamma(0)} = \rho^s$$

What happens if $\rho = 1$? Would it be stationary then?

We really do not believe that many economic series can be explosive so we usually ignore this possibility.

AR(1) example

Example

$$
\begin{array}{lll}
Y_t = \rho Y_{t-1} + \varepsilon_t & \rho = 0.4 & \rho = 0.9 \\
 & \rho(1) = 0.4 & \rho(1) = 0.9 \\
 & \rho(2) = 0.16 & \rho(2) = 0.81 \\
 & \rho(3) = 0.064 & \rho(3) = 0.729
\end{array}
$$

- For small coefficient the ACF between now and the past becomes very small fast, for large values (close to 1) the correlation (e.g. effect of a shock) stay large for a long time. We say that the coefficient is a measure of the persistence in the data.
- Lots of discussion on what the size of the largest root for most macro data is!

	MA	AR
ACF	drop to 0 after q lags	decays to zero
PACF	decays to zero	drops to zero after p lags

ARMA

$$y_t = \rho y_{t-1} + \varepsilon_t$$

shock

- unpredictable
  ||
  random close to zero

Shock $=1$

$y_{t-1} = 0$

$y_t = \varepsilon_t = 1$

$y_{t+1} = \rho \cdot y_t = \rho$

$y_{t+2} = \rho^2$

$y_{t+3} = \rho^3$

<u>shock today only</u>

$\varepsilon_t = 1$

$\varepsilon_{t+1} = 0$

$= 0$

$= 0$

ACF and PACF for AR

Autocorrelation and partial autocorrelation for AR processes have a very specific shape.

- PACF $\rightarrow$ it is different than zero for $j \leq p$ and zero otherwise
- ACF $\rightarrow$ decays or oscillates slowly

Importance of the root of AR(1) process

You can see why the point at which $\phi = 1$ is of particular interest

- The series in not stationary
- The effect of shocks to the economy never disappears
- The Wold theorem may not apply
- The variance can be infinite
- I cannot construct sensible forecasts (more later)
- All the asymptotic theory breaks down
- Tests do not have the usual distribution

It is important to distinguish statistically when "the root" is equal to one (thus all these papers out there)

AR(2)

don't need to know math

$$\left(1-\rho_1 L-\rho_2 L^2\right) Y_t = c+\varepsilon_t$$

$$(1-aL)(1-bL)Y_t = c+\varepsilon_t$$

$$Y_t = \frac{1}{1-aL}\frac{1}{1-bL}(c+\varepsilon_t) = \frac{1}{1-a}\frac{1}{1-b}c + \left(\sum_{i=0}^{\infty} a^i L^i\right)\left(\sum_{j=0}^{\infty} b^j L^j\right)\varepsilon_t$$

The conditions we need for convergence are $|a|<1$ and $|b|<1$

This is equivalent to saying that the roots of $1-\rho_1 z-\rho_2 z^2=0$ are larger than one. Often you will see statements like "the roots are outside the unit circle". This condition assures that $\sum|\psi_j|<\infty$ in the MA representation which assures stationarity.

ARMA

ARMA(p, q) model are a mix of the two

$$Y_t = c + \rho_1 Y_{t-1} + \ldots + \rho_p Y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + .. + \theta_q \varepsilon_{t-q}$$

ACF $\rightarrow$ decays or oscillates slowly PACF $\rightarrow$ decays or oscillates slowly

- Look at handout on correlogram and you can see. Box and Jenkins approach.
- We do not use that much anymore. We use Information Criteria to choose between models. But, in my opinion, it is a good practice to have a look. If ACF and PACF show correlation your need to add lags!

Goal: get to no serial correlation in residuals

ARMA to MA

This is true in general: Any $AR(p)$ with *roots outside the unit circle* has an MA representation. These processes are called stationary (because there is a weakly stationary version of them).

Any MA process with roots outside unit circle can be written as $AR(\infty)$, such processes called invertible. If $y_t = b(L)e_t$ is an *invertible* MA process, then $e_t = b(L)^{-1}y_t$. That is, the "errors" are laying in a space of observations and can be recovered from y 's (another name for this: *errors are fundamental*).

Invertibility

Consider a MA(1) process

$$Y_t - \mu = (1 + \theta L)\varepsilon_t$$

$$E\left(\varepsilon_t \varepsilon_s\right) = \begin{cases} \sigma^2 & \text{if } t = s \\ 0 & \text{otherwise} \end{cases}$$

If $|\theta| < 1$

$$(1 + \theta L)^{-1}\left(Y_t - \mu\right) = \varepsilon_t$$

$$\left(1 - \theta L + \theta^2 L^2 - \theta^3 L^3 + \ldots.\right)\left(Y_t - \mu\right) = \varepsilon_t \quad AR(\infty)$$

If a $MA(1)$ can be written as a AR(∞), then the process if invertible. What this means is that we can write the present as function of the past in a way that the more distance past is less important.

Estimation

Once we have decided which model may be appropriate, how do we estimate it?

How to find the best model:

bypass for now

- Make sure the data is stationary first (i,e detrend first or include trend)
- Look at correlogram (of data or residuals if you have trend and ~~stationarity~~) Seasonality
- Estimate and look at individual significance (large to small versus small to large)
- Look at Information Criteria
- Look at graph and correlogram of the residuals
- Stop when you have exploited any forecastability (ie. when residuals are completely random)

if it's flat, you're done

smaller the better

Tuesday we will do
from top to bottom whole thing

real data can be explained well of autoregressive modell

$$y_t = \phi y_{t-1} + \varepsilon_t$$

$$y_t = \phi y_{t-1} + \phi^2 y_{t-2} + \phi^3 y_{t-3}$$

$$y_t = \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-1} + \phi^3 \varepsilon_{t-3} +$$

$$y_t = \varepsilon_t + \theta \varepsilon_{t-1}$$

$$MA(1) \Longleftrightarrow AR(\infty)$$

$$AR(1) = MA(\infty)$$

People don't like moving model average, good place to start is AR

	AIC	BIC
AR(1)	1384	1397
AR(2)	1386	1403
ARMA(1,1)	1386	1402
ARMA(2,2)	1385	1410

ARMA(2,1) 1383 1400